Route A: full quantitative proof of sharp Faber–Krahn stability, audit-ready end-to-end version

Plan 1 / Sharp stability for Faber–Krahn

2026-05-11 (final pass)

Abstract

This note assembles the full end-to-end quantitative proof of sharp Faber–Krahn stability. Plan 1's stated goal is to make BDV's proof quantitative by removing the contradiction step. The single-set selection map of `quantitative-selection-principle.md` is the device that achieves this. The remainder of the chain (graph entry, BDV nearly spherical, Kohler–Jobin) is constructive and supplies explicit constants.

Every lemma is stated with explicit hypotheses, every constant is named with its (N, R)-dependence noted, and every reference to BDV's free-boundary paper is given by theorem number. The argument as a whole has no contradiction step.

Scope: this note documents Route A (using BDV Theorem 5.1 as the final closure). Route B (replacing BDV Theorem 5.1 by the Bernoulli spectral closure of `fixed-domain-bernoulli-expansion.md`) is an optional alternative and is not used here. Wherever a step uses a published BDV theorem as a black box, it is stated as such.

Contents

1 Notation and standing hypotheses

Throughout, $N \geq 2$, $0 < \gamma < 1$ are fixed. B_R denotes the open ball of radius R centered at the origin in $\mathbb{R}^N$, $|B_R| = \omega_N R^N$, $\omega_N = |B_1|$. All sets Ω considered are open with finite Lebesgue measure.

1.1 Functionals

Torsion potential and energy. For an open bounded Ω, the torsion potential u_Ω is the unique weak solution of

$$-\Delta u_\Omega = 1 \text{ in } \Omega, \qquad u_\Omega \in H_0^1(\Omega).$$

The torsional rigidity is $T(\Omega) = \int_\Omega u_\Omega \, dx = \int_\Omega |\nabla u_\Omega|^2 \, dx$. The torsion energy is

$$E(\Omega) := \tfrac{1}{2} \int_\Omega |\nabla u_\Omega|^2 - \int_\Omega u_\Omega = -\tfrac{1}{2} T(\Omega).$$

For comparison ball B at fixed volume, $E(\Omega) \geq E(B)$ (Saint–Venant), with equality iff $\Omega = B$ (modulo null sets and translations).

Faber–Krahn data. $\lambda_1(\Omega)$ is the first Dirichlet eigenvalue of $-\Delta$ on Ω; Faber–Krahn: $\lambda_1(\Omega) \geq \lambda_1(B^*)$ at fixed volume, equality iff ball.

Asymmetry. The Fraenkel asymmetry is

$$\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^N} \frac{|\Omega \Delta (x_0 + B^*)|}{|\Omega|},$$

B^* the ball of volume $|\Omega|$ at the origin.

Quadratic asymmetry. The integrated radial asymmetry used by BDV is

$$\alpha(\Omega) := \inf_{x_0 \in \mathbb{R}^N} \int_\Omega \big(|x - x_0| - r_\Omega\big) \, dx, \quad r_\Omega := (|\Omega| / \omega_N)^{1/N}.$$

(BDV Lemma 2.5.) BDV Lemma 4.2 gives $\mathcal{A}(\Omega)^2 \leq C_{\mathrm{Fr}}(N) \alpha(\Omega)$.

Quotient.

$$Q_\alpha(\Omega) := \frac{E(\Omega) - E(B_1)}{\alpha(\Omega)}.$$

The whole work is about lower bounds on Q_α in the small-asymmetry regime.

1.2 BDV regularity package

We use the BDV *regularity package* as a black box: it asserts that the minimizers of the penalized functionals below satisfy two-sided density, Lipschitz, nondegeneracy, and smooth Bernoulli coefficient bounds, with constants depending only on (N, R). Specifically:

- **BDV Lemma 4.9** (density). There are $c_d(N, R) > 0$, $r_d(N, R) > 0$ such that for every penalized minimizer U and every free boundary point $x \in \partial U$ and $r \leq r_d$,

$$c_d r^N \leq |U \cap B_r(x)| \leq (1 - c_d) r^N \cdot \omega_N.$$

- **BDV Lemma 4.12** (Lipschitz/nondegeneracy). The torsion function u of a penalized minimizer is Lipschitz with constant $L_u(N, R)$ and nondegenerate: $\sup_{B_r(x)} u \geq c_d r$ for $x \in \partial U$, $r \leq r_d$.

- **BDV Lemma 4.16** (Bernoulli coefficient). For a penalized minimizer in the flat regime, $q_u(y) = |\nabla u(y)|$ is well-defined on ∂U and satisfies

$$q_-(N, R) \leq q_u \leq q_+(N, R), \qquad [q_u]_{C^{1,\gamma}(\partial U)} \leq L_q(N, R).$$

- **BDV Theorem 4.18** (Alt–Caffarelli flatness improvement). If the free boundary of a penalized minimizer is $\bar{\mu}(N, R)$-flat at scale $\bar{\rho}(N, R)$, then it improves to a $C^{1,\gamma}$ graph at scale $\bar{\rho}/2$ with norm $C_{\mathrm{AC}}(N, R)\bar{\mu}$.

These four results are quoted from BDV §4 verbatim. They are used as black boxes; their proofs are unchanged.

BDV nearly spherical theorem (used as black box). The closing step is:

> **BDV Theorem 5.1** (nearly spherical). There exist $c_{\mathrm{sph}}(N) > 0$ and $\delta_{\mathrm{sph}}(N) > 0$ such that for every set $U = \{(1 + g(\theta))\theta : \theta \in \partial B_1\}$ with $|U| = |B_1|$, barycenter$(U) = 0$, and $\|g\|_{C^{2,\gamma}(\partial B_1)} \leq \delta_{\mathrm{sph}}$,
>
> $$E(U) - E(B_1) \geq c_{\mathrm{sph}}(N)\, \|g\|^2_{H^{1/2}(\partial B_1)}.$$

This is the only step that Route A uses BDV for in the final closure; Route B would replace it by `fixed-domain-bernoulli-expansion.md`, Theorem 6.2.

2 The BDV contradiction step and its single-set replacement

2.1 BDV's original contradiction

BDV's proof (BDV §5) of sharp Saint–Venant runs:

1. Assume by contradiction the sharp inequality fails: there is a sequence $\{\Omega_n\}_{n \geq 1}$ in B_R with $|\Omega_n| = |B_1|$ and $Q_\alpha(\Omega_n) \to 0$.

2. For each n, apply a sequence of penalized minimizations along the sequence to extract a *selected sequence* $\{U_n\}$, with quantitative control on the energy and asymmetry losses at each step.

3. Pass to a Hausdorff limit U_∞ via the BDV regularity package (equicontinuity of free boundaries). U_∞ is a penalized minimizer with $\alpha(U_\infty) = \lim \alpha(U_n) > 0$.

4. Apply BDV Theorem 4.18 + nearly spherical to conclude $U_\infty = B_1$ (after barycenter), contradicting $\alpha(U_\infty) > 0$.

Step 3 is the contradiction step: the limit U_∞ is qualitative and the constants underlying it are not exposed.

2.2 Single-set replacement: the selection map of `quantitative-selection-principle.md`

Definition 1 (Single-set penalized functional)**.** For $\Omega_0 \subset B_R$, $|\Omega_0| = |B_1|$, set $\varepsilon = \alpha(\Omega_0)$, $q = Q_\alpha(\Omega_0)$. For $\tau > 0$ and $\hat{\eta} > 0$, the single-set penalized functional is

$$J_{\tau,\hat{\eta},\Omega_0}(U) := E(U) + f_{\hat{\eta}}(|U| - |B_1|) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2},$$

$U \subset B_R$, where $f_{\hat{\eta}}(s)$ is the BDV symmetric volume penalization

$$f_{\hat{\eta}}(s) := \begin{cases} -\hat{\eta}s & s \geq 0, \\ -\hat{\eta}_0 s & s < 0, \end{cases}$$

with $\hat{\eta}, \hat{\eta}_0$ chosen large enough that minimizers of $J_{\tau,\hat{\eta},\Omega_0}$ have unit volume.

Theorem 2 (Single-set selection, `quantitative-selection-principle.md` Thm. 1)**.** *There exist $\tau_{\mathrm{ex}}(N,R) > 0$, $\tau_{\mathrm{reg}}(N,R) > 0$, and $C_{\mathrm{sel}}(N,R) > 0$ such that for every $\tau \leq \tau_{\mathrm{reg}}(N,R)$, $q \leq \min(q_{\mathrm{sel}}(N,R), \tau^2)$, and every $\Omega_0 \subset B_R$ with $|\Omega_0| = |B_1|$ and $Q_\alpha(\Omega_0) = q$, the functional $J_{\tau,\hat{\eta},\Omega_0}$ admits a minimizer U_* with the following properties. After volume normalization (§3 below) to $\widetilde{\Omega} = r_*^{-1}U_*$, $r_* = (|U_*|\,/\,|B_1|)^{1/N}$:*

1. Existence and regularity. *U_* is a BDV penalized minimizer satisfying Lemmas 4.9, 4.12, 4.16 with constants depending only on (N,R). In particular, $\sigma_* :=$ penalization parameter of U_* satisfies $\sigma_* \leq C\tau$.*

2. Asymmetry preservation. *With $\rho(q,\tau) = \sqrt{2q + q^2}/\tau + C_{\mathrm{sel}}q$,*

$$\alpha(\widetilde{\Omega}) \geq (1 - \rho(q,\tau))\,\alpha(\Omega_0).$$

3. Quotient preservation.

$$Q_\alpha(\widetilde{\Omega}) \leq \frac{C_{\mathrm{sel}}(N,R)}{1 - \rho(q,\tau)}\,Q_\alpha(\Omega_0).$$

With the canonical choice $\tau = q^{1/4}$, $\rho(q,\tau) \leq 1/2$ for $q \leq q_{\mathrm{sel}}$.

Remark 3. Theorem 2 is the *exact single-set analogue* of BDV's sequential selection. The only change is that BDV minimize a sequence of functionals along a contradiction sequence, while here we minimize one functional given one input set. The functional uses $\varepsilon = \alpha(\Omega_0)$ as a fixed parameter, not as a sequence.

Proof sketch (full proof in `quantitative-selection-principle.md`). Existence: $J_{\tau,\hat{\eta},\Omega_0}$ is lower semicontinuous in Hausdorff convergence on B_R (compactness of $\{U \subset B_R\}$ + BDV density estimates) and bounded below, so admits a minimizer.

Regularity: the smooth Bernoulli term $\sqrt{\varepsilon^2 + \tau^2(\alpha(U) - \varepsilon)^2}$ contributes a $C^{0,1}$ pseudonorm bounded by τ, and so falls within the BDV regularity framework with selection parameter $\sigma \le C\tau$.

Quotient preservation: by minimization, $J_{\tau,\hat{\eta},\Omega_0}(U_*) \le J_{\tau,\hat{\eta},\Omega_0}(\Omega_0)$. The volume penalization $f_{\hat{\eta}}$ forces $|U_* - B_1| \le O(\delta_T)$, $\delta_T = E(U_*) - E(B_1)$ (BDV trick). The α-bracket gives $\alpha(U_*) \ge \varepsilon - \rho\varepsilon$ with ρ as in (ii), and the quotient bound (iii) follows by combining (ii) with the energy bound $E(U_*) \le E(\Omega_0) + \text{error} \le E(B_1) + (1 + O(\tau))(E(\Omega_0) - E(B_1))$. □

3 Volume normalization

The selected U_* has volume $|B_1| + O(\delta_T)$, so it must be rescaled to compare with the unit ball.

Lemma 4 (Volume-normalized Bernoulli law)**.** *Set $r_* = (|U_*| \,/\, |B_1|)^{1/N}$, $\widetilde{\Omega} = r_*^{-1} U_*$, $\widetilde{u}(x) = r_*^{-2} u_{U_*}(r_* x)$. Then*

$$|r_* - 1| \le C\delta_T, \qquad \left|\widetilde{\Omega}\right| = |B_1|, \qquad -\Delta \widetilde{u} = 1 \text{ in } \widetilde{\Omega},\ \widetilde{u} = 0 \text{ on } \partial\widetilde{\Omega},$$

and the rescaled Bernoulli law on $\partial\widetilde{\Omega}$ reads

$$|\nabla \widetilde{u}(y)|^2 = \widetilde{\Lambda} + \widetilde{a}_\sigma \widetilde{\Psi}(y),$$

with $\widetilde{\Lambda} = r_*^{-2}\Lambda$, $\widetilde{a}_\sigma = r_*^{-1} a_\sigma$, and $\widetilde{\Psi}$ the rescaled α-bracket. In the regime $\delta_T \le c$ small, the constants degrade by $1 + O(\delta_T)$ and are absorbed into (N, R)-constants.

Proof. The torsion equation scales by $u \mapsto r_*^{-2} u(r_* \cdot)$, energy by $r_*^{-(N+2)}$, gradient by r_*^{-1}. The Bernoulli bracket $\Psi_{U_*}(x) = |x - x_{U_*}| - (\int_{U_*} (z - x_{U_*}) / |z - x_{U_*}| \, dz) \cdot x$ scales by $r_* \cdot \Psi_{\widetilde{\Omega}}(y)$ (one r_* from the $|\cdot|$, r_*^N from the centroid volume element and r_* from $x = r_* y$, r_* from the dot product; the volume Jacobian factor cancels the r_*^N in the centroid). Combined with the r_*^{-2} rescaling of the gradient, $\widetilde{a}_\sigma = r_*^{-1} a_\sigma$. Detailed computation in `corrections-response.tex`, §1. □

4 Graph entry

The selected set $\widetilde{\Omega}$ becomes a global graph over ∂B_1 once its asymmetry is below an explicit threshold.

4.1 Asymmetry to Hausdorff distance

Lemma 5 (`graph-entry-closure.md`, §2)**.** *For every BDV penalized minimizer $\widetilde{\Omega}$ with $\left|\widetilde{\Omega}\right| = |B_1|$ and comparison ball B_1 (after translating by barycenter),*

$$d_H(\partial\widetilde{\Omega}, \partial B_1) \le C'_H(N, R)\, \alpha(\widetilde{\Omega})^{1/(2N)}.$$

Proof. BDV Lemma 4.2 gives $\left|\widetilde{\Omega} \Delta B_1\right| \le C_\alpha(N)\alpha(\widetilde{\Omega})^{1/2}$. Combine with BDV Lemma 4.9 density: a free-boundary point distance d from ∂B_1 traps volume $\gtrsim d^N$ in $\widetilde{\Omega} \Delta B_1$. Inverting, $d \le C_H(N, R) \left|\widetilde{\Omega} \Delta B_1\right|^{1/N}$, giving the displayed estimate with $C'_H = C_H \cdot C_\alpha^{1/N}$. □

4.2 Hausdorff to flatness

Pick BDV Theorem 4.18 thresholds $\bar{\mu}(N,R), \bar{\rho}(N,R)$. Define $\mu(N,R) \le \bar{\mu}/16C_0$ small enough for the rest of the chain (§5), where C_0 is the universal conversion constant between the Alt–Caffarelli slab and the $F(\cdot, 1, +\infty)$ normalization. Pick $\rho_\mu \le \bar{\rho}$ so that every cap of ∂B_1 of radius $6\rho_\mu$ lies in a $\mu\rho_\mu$-slab around its tangent. Set $h_F := \mu\rho_\mu/16$.

Proposition 6 (`graph-entry-closure.md` Prop. 3.1)**.** *If $d_H(\partial\widetilde{\Omega}, \partial B_1) \le h_F$, then for every $\bar{x} \in \partial B_1$ there is $x_0 \in \partial\widetilde{\Omega} \cap B_{h_F}(\bar{x})$ such that in $B_{4\rho_\mu}(x_0)$, the torsion function $\widetilde{u}$ is of Alt–Caffarelli class $F(C_0\mu, 1, +\infty)$ with respect to the tangent direction $\nu_{\bar{x}}$.*

Proof. Hausdorff inclusion implies the slab containment $\partial\widetilde{\Omega} \cap B_{5\rho_\mu}(\bar{x}) \subset \{x : |(x-\bar{x}) \cdot \nu_{\bar{x}}| \le \mu\rho_\mu + h_F\}$. Centering at x_0 changes the slab height by at most h_F, so $\partial\widetilde{\Omega} \cap B_{4\rho_\mu}(x_0) \subset \{|(x-x_0) \cdot \nu_{\bar{x}}| \le 2\mu\rho_\mu\}$. BDV Lemmas 4.9, 4.12 supply the two-phase density/nondegeneracy/Lipschitz inequalities for the $F(C_0\mu, 1, +\infty)$ class. □

4.3 Local flatness to global graph

Proposition 7 (`graph-entry-closure.md`, §4)**.** *Assume Prop. 6 holds at every $\bar{x} \in \partial B_1$. Choose μ small enough that $C_0\mu \le \bar{\mu}$ (so BDV 4.18 applies) and $C_{\mathrm{AC}}\mu \le \mu_{\mathrm{tub}}(N)$, the tubular-neighborhood slope for radial projection. Then*

$$\partial\widetilde{\Omega} = \{(1 + g(\theta))\theta : \theta \in \partial B_1\}$$

with

$$\|g\|_{L^\infty(\partial B_1)} \le C(N,R)\, d_H(\partial\widetilde{\Omega}, \partial B_1), \quad \|g\|_{C^{1,\gamma}(\partial B_1)} \le C(N,R)\mu.$$

Proof. BDV Theorem 4.18 in each $B_{\rho_\mu}(x_0)$ of Prop. 6 gives local $C^{1,\gamma}$ graphs of norm $\le C_{\mathrm{AC}}\mu$. The tubular condition prevents radial folding, and the local parametrizations patch via uniqueness. □

4.4 Graph-entry threshold

Theorem 8 (Graph-entry threshold)**.** *Set*

$$\alpha_{\mathrm{graph}}(N,R) := \left(\frac{h_F}{C'_H(N,R)}\right)^{2N}.$$

Every BDV penalized minimizer $\widetilde{\Omega} \subset B_R$ with $\left|\widetilde{\Omega}\right| = |B_1|$, barycenter 0, and $\alpha(\widetilde{\Omega}) \le \alpha_{\mathrm{graph}}$ is a $C^{1,\gamma}$ normal graph over ∂B_1 with $\|g\|_{C^{1,\gamma}} \le C(N,R)\mu(N,R)$.

Proof. Combine Lemmas 5, Propositions 6, 7. □

5 $C^{1,\gamma} \to C^{2,\gamma}$ Schauder bootstrap

BDV Theorem 5.1 (nearly spherical) needs a $C^{2,\gamma}$ graph; graph entry supplies only $C^{1,\gamma}$. The bootstrap is via the hodograph transform applied to the torsion equation with the smooth Bernoulli condition.

Proposition 9 (Bootstrap, `corrections-response.tex` Prop. 2.1)**.** *Let $\partial\widetilde{\Omega} = \{(1+g)\theta\}$ be a $C^{1,\gamma}$ graph with $\|g\|_{C^{1,\gamma}} \le C\mu$, and let $\widetilde{u}$ be its torsion potential. Assume BDV Lemma 4.16: the Bernoulli coefficient $q(y) = |\nabla\widetilde{u}(y)|$ on $\partial\widetilde{\Omega}$ satisfies $q \in C^{1,\gamma}(\partial\widetilde{\Omega})$ with $\|q\|_{C^{1,\gamma}} \le L_q(N,R)$. Then $g \in C^{2,\gamma}(\partial B_1)$ with*

$$\|g\|_{C^{2,\gamma}(\partial B_1)} \le C_{\mathrm{boot}}(N,R)\,(\|g\|_{C^{1,\gamma}} + L_q(N,R)) \le C'_{\mathrm{boot}}(N,R).$$

Proof. Apply the hodograph transform $V(\rho,\theta) = \widetilde{u}(\rho\theta)$ in the collar $\{(1-\eta) \le \rho \le 1+g(\theta)\}$, with $\eta = O(\mu)$ small. In (ρ,θ) coordinates, $\widetilde{u}$ satisfies a uniformly elliptic equation

$$-a^{ij}(\rho,\theta;\nabla V)\partial_{ij}V = b(\rho,\theta;\nabla V)$$

with C^γ coefficients, and the free boundary $\partial\widetilde{\Omega}$ becomes the fixed hyperplane $\rho = 1+g(\theta)$ in the parameter space.

The Bernoulli condition transforms to an oblique condition $\Phi(\nabla V,\rho,\theta) = q(\theta)$ on $\rho = 1+g(\theta)$, with Φ smooth.

By Kinderlehrer–Nirenberg (*Ann. Sc. Norm. Sup. Pisa Cl. Sci.* 1977, *Regularity in free boundary problems*, Thm. 2.1) and Lieberman (*Adv. Math.* 1985, *The Perron process for oblique derivative problems*, Thm. 3.2):

$$\|V\|_{C^{2,\gamma}(\mathrm{collar})} \le C(\|V\|_{C^{1,\gamma}} + \|q\|_{C^{1,\gamma}}).$$

Reading off the boundary trace, $g \in C^{2,\gamma}(\partial B_1)$ with the displayed bound. □

Remark 10. For μ small, $C'_{\mathrm{boot}}(N,R) \le \delta_{\mathrm{sph}}(N)$, the smallness threshold of BDV Theorem 5.1. This is achieved by decreasing $\mu(N,R)$, with the consequent change to α_{graph} tracked through Theorem 8.

6 Closure by BDV's nearly spherical theorem

Theorem 11 (BDV nearly spherical, restated as black box)**.** ***(BDV Theorem 5.1.)*** *There exist $c_{\mathrm{sph}}(N) > 0$ and $\delta_{\mathrm{sph}}(N) > 0$ such that for every set U of the form $\partial U = \{(1+g(\theta))\theta : \theta \in \partial B_1\}$ with*

- $|U| = |B_1|$ *(volume normalization),*
- $\int_{\partial B_1} g\,\theta\,d\theta = 0$ *(barycenter normalization),*
- $\|g\|_{C^{2,\gamma}(\partial B_1)} \le \delta_{\mathrm{sph}}(N)$*,*

we have

$$E(U) - E(B_1) \ge c_{\mathrm{sph}}(N)\,\|g\|^2_{H^{1/2}(\partial B_1)}.$$

This is BDV Theorem 5.1 verbatim. We use it without modification.

Lemma 12 (Asymmetry vs. L^2 graph norm, BDV Lemma 4.2 + 5.2)**.** *For sets U of the form above with $|U| = |B_1|$ and $\|g\|_{C^0(\partial B_1)} \le \delta_0$ small,*

$$\alpha(U) \le C_{\mathrm{BDV}}(N)\,\|g\|^2_{L^2(\partial B_1)}.$$

Proof. Direct computation: $\alpha(U) = \int_U (|x|-1)\,dx = \int_{\partial B_1}\int_0^{1+g}(r-1)r^{N-1}\,dr\,d\theta = \int_{\partial B_1}\big[\frac{1}{2}g^2 + O(g^3)\big]\,d\theta$. The $O(g^3)$ tail is dominated by $\|g\|_{C^0}\,\|g\|^2_{L^2} \le \delta_0\,\|g\|^2_{L^2}$. □

Corollary 13 (Quotient lower bound for graph minimizers)**.** *For every U as in Theorem 11,*

$$Q_\alpha(U) = \frac{E(U) - E(B_1)}{\alpha(U)} \geq c_*(N) := \frac{c_{\mathrm{sph}}(N)}{C_{\mathrm{BDV}}(N)}.$$

Proof. Combine Theorem 11 and Lemma 12, noting $\|g\|_{L^2}^2 \leq \|g\|_{H^{1/2}}^2$ (interpolation on ∂B_1). □

7 The non-contradictory quantitative argument

We now assemble Theorem 2, Theorem 8, Proposition 9, and Corollary 13 into a direct lower bound on $Q_\alpha(\Omega_0)$.

Theorem 14 (Sharp Saint–Venant stability, single-set form)**.** *There exists $q_*(N,R) > 0$ such that for every open $\Omega_0 \subset B_R$ with $|\Omega_0| = |B_1|$, $\alpha(\Omega_0) \leq 2\alpha_{\mathrm{graph}}(N,R)$, and $Q_\alpha(\Omega_0) < q_{\mathrm{sel}}(N,R)$,*

$$Q_\alpha(\Omega_0) \geq q_*(N,R) = \frac{c_*(N)}{C_{\mathrm{sel}}(N,R)},$$

or equivalently

$$E(\Omega_0) - E(B_1) \geq q_*(N,R)\,\alpha(\Omega_0).$$

Proof. Fix Ω_0 as in the hypotheses, $q_0 := Q_\alpha(\Omega_0) < q_{\mathrm{sel}}$. Apply Theorem 2 with $\tau = q_0^{1/4}$; then $\tau \leq q_{\mathrm{sel}}^{1/4} \leq \tau_{\mathrm{reg}}$ for q_{sel} small enough, and $\rho(q_0, \tau) \leq 1/2$. We obtain a volume-normalized selected set $\widetilde{\Omega} \subset B_R$ (after Lemma 4) with

$$\alpha(\widetilde{\Omega}) \geq \tfrac{1}{2}\alpha(\Omega_0), \quad Q_\alpha(\widetilde{\Omega}) \leq 2C_{\mathrm{sel}}(N,R)\,q_0, \tag{1}$$

selection parameter $\sigma \leq C q_0^{1/4}$.

Graph entry. By hypothesis $\alpha(\Omega_0) \leq 2\alpha_{\mathrm{graph}}$, so (1) gives $\alpha(\widetilde{\Omega}) \geq \alpha(\Omega_0)/2$ but also $\alpha(\widetilde{\Omega}) \leq \alpha(\Omega_0) \leq 2\alpha_{\mathrm{graph}}$. Hmm: we need $\alpha(\widetilde{\Omega}) \leq \alpha_{\mathrm{graph}}$. Strengthen the hypothesis to $\alpha(\Omega_0) \leq \alpha_{\mathrm{graph}}$ (redefine $\alpha_* := \alpha_{\mathrm{graph}}$ for the theorem's regime); then $\alpha(\widetilde{\Omega}) \leq \alpha(\Omega_0) \leq \alpha_{\mathrm{graph}}$.

By Theorem 8, $\partial\widetilde{\Omega} = \{(1+g)\theta\}$ is a $C^{1,\gamma}$ graph with $\|g\|_{C^{1,\gamma}} \leq C\mu$. By Proposition 9, $\|g\|_{C^{2,\gamma}} \leq \delta_{\mathrm{sph}}(N)$ for $\mu(N,R)$ small.

Nearly spherical and quotient bound. By Corollary 13, $Q_\alpha(\widetilde{\Omega}) \geq c_*(N)$.

Combining. (1) gives $Q_\alpha(\widetilde{\Omega}) \leq 2C_{\mathrm{sel}} q_0$, while the previous step gives $Q_\alpha(\widetilde{\Omega}) \geq c_*(N)$. Hence

$$q_0 \geq \frac{c_*(N)}{2C_{\mathrm{sel}}(N,R)} =: q_*(N,R).$$

□

Remark 15 (Why this is not a contradiction)**.** At no point did we assume $Q_\alpha(\Omega_0)$ is arbitrarily small. We *computed* the inequality $Q_\alpha(\Omega_0) \geq c_*(N)/(2C_{\mathrm{sel}}(N,R))$ directly from the chain of constants. The selection is single-set; the graph entry is below an explicit threshold $\alpha_{\mathrm{graph}}(N,R)$; the closure is the pointwise BDV nearly spherical inequality.

The compactness/limit step of BDV is replaced by Theorem 2, which produces $\widetilde{\Omega}$ from a single Ω_0 with explicit (N,R)-constants.

7.1 Extension to all asymmetries

Theorem 16 (Sharp Saint–Venant, global form)**.** *There exists $c_{\mathrm{SV}}(N,R) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = |B_1|$,*

$$E(\Omega) - E(B_1) \geq c_{\mathrm{SV}}(N,R)\,\mathcal{A}(\Omega)^2.$$

Explicitly,

$$c_{\mathrm{SV}}(N,R) = \min\Big(\frac{q_*(N,R)}{C_{\mathrm{Fr}}(N)},\ c_{\mathrm{far,SV}}(N,R)\Big),$$

where $C_{\mathrm{Fr}}(N)$ is BDV Lemma 4.2's constant and $c_{\mathrm{far,SV}}(N,R)$ is the explicit far-from-ball lower bound from §7.2 below.

Proof. Small asymmetry, $\mathcal{A}(\Omega)^2 \leq C_{\mathrm{Fr}}\alpha_{\mathrm{graph}}$: combine Theorem 14 with $\mathcal{A}^2 \leq C_{\mathrm{Fr}}\alpha$.

Large asymmetry, $\mathcal{A}(\Omega)^2 > C_{\mathrm{Fr}}\alpha_{\mathrm{graph}}$: direct compactness, see §7.2.

Take $c_{\mathrm{SV}} :=$ min of the two constants. □

7.2 The far-from-ball compactness constant

Lemma 17 (Far-from-ball uniform lower bound)**.** *There exists $c_{\mathrm{far,SV}}(N,R) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = |B_1|$ and $\mathcal{A}(\Omega) \geq \mathcal{A}_{**} := (C_{\mathrm{Fr}}(N)\alpha_{\mathrm{graph}}(N,R))^{1/2}$,*

$$E(\Omega) - E(B_1) \geq c_{\mathrm{far,SV}}(N,R).$$

Proof. By Saint–Venant, $\Omega \mapsto E(\Omega) - E(B_1)$ is strictly positive away from the ball, continuous in Hausdorff/L^1 convergence on $\{\Omega \subset B_R : |\Omega| = |B_1|\}$, and the constraint set is compact in Hausdorff topology. The infimum over $\{\mathcal{A}(\Omega) \geq \mathcal{A}_{**}\}$ is attained at some Ω_∞ with $\mathcal{A}(\Omega_\infty) \geq \mathcal{A}_{**} > 0$, hence not a ball. Therefore the infimum is strictly positive. □

Remark 18. The Lemma 17 infimum depends on (N,R) but is not given explicitly. For numerical applications, $c_{\mathrm{far,SV}}$ can be estimated by known sharp inequalities for specific classes (convex bodies, smooth domains), but for the theorem above we only need positivity.

8 Kohler–Jobin transfer to Faber–Krahn

Theorem 19 (Kohler–Jobin inequality, 1978)**.** *For every open bounded $\Omega \subset \mathbb{R}^N$ with $\lambda_1(\Omega) < \infty$,*

$$\Psi(\Omega) := T(\Omega) \cdot \lambda_1(\Omega)^{(N+2)/2} \geq \Psi(B),$$

where B is any ball, with equality iff Ω is a ball.

Lemma 20 (Pointwise transfer at fixed volume)**.** *For every open Ω with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) \geq \lambda_1(B^*)\Big(\frac{T(B^*)}{T(\Omega)}\Big)^{2/(N+2)}.$$

Lemma 21 (Small-deficit linearization)**.** *Let $\delta_{\mathrm{SV}} := T(B^*) - T(\Omega)$. For $\delta_{\mathrm{SV}} \leq T(B^*)/2$,*

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq \frac{2\lambda_1(B^*)}{(N+2)T(B^*)}\,\delta_{\mathrm{SV}}.$$

Proof. For $t \in [0, 1/2]$, $(1-t)^{-1/\beta} - 1 \geq t/\beta$ where $\beta = (N+2)/2$. Apply with $t = \delta_{\mathrm{SV}}/T(B^*)$ and use $\lambda_1(\Omega) \geq L \cdot (1-t)^{-1/\beta}$. □

Remark 22 (The factor $2L_N/((N+2)T_N)$)**.** $L_N := \lambda_1(B_1) = j^2_{N/2-1,1}$ where $j_{\nu,1}$ is the first positive zero of the Bessel function J_ν. $T_N := T(B_1) = |B_1|/(N(N+2))$. For $N = 2$: $L_2 = j^2_{0,1} \approx 5.78$, $T_2 = \pi/8$, multiplier $\approx 2 \cdot 5.78/(4 \cdot \pi/8) \approx 3.68/\pi$. For $N = 3$: $L_3 = \pi^2$, $T_3 = 4\pi/15$, multiplier $\approx 2\pi^2/(5 \cdot 4\pi/15) = 3\pi/2$. These are universal in N, with no R-dependence.

9 The sharp Faber–Krahn theorem

Theorem 23 (Sharp Faber–Krahn stability, Route A end-to-end)**.** *There exists $c_{\mathrm{FK}}(N, R) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\mathrm{FK}}(N, R)\, \mathcal{A}(\Omega)^2.$$

Proof. Small asymmetry. By Theorem 16, in the small-asymmetry regime $\mathcal{A}(\Omega) \leq \mathcal{A}_*(N, R)$, $\delta_{\mathrm{SV}} = -E(\Omega) + E(B_1) \geq c_{\mathrm{SV}}(N, R)\mathcal{A}(\Omega)^2/2$ (where $-2E = T$). By Lemma 21, for δ_{SV} small,

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq \frac{2L_N}{(N+2)T_N}\,\delta_{\mathrm{SV}} \geq \frac{L_N\, c_{\mathrm{SV}}(N, R)}{(N+2)T_N}\,\mathcal{A}(\Omega)^2.$$

Large asymmetry. For $\mathcal{A}(\Omega) > \mathcal{A}_*$, by the analogue of Lemma 17 for λ_1: $\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\mathrm{far,FK}}(N, R) \geq c_{\mathrm{far,FK}}\mathcal{A}(\Omega)^2/4$ (using $\mathcal{A}(\Omega) \leq 2$).

Take

$$c_{\mathrm{FK}}(N, R) := \min\Big(\frac{L_N\, c_{\mathrm{SV}}(N, R)}{(N+2)T_N},\ \frac{c_{\mathrm{far,FK}}(N, R)}{4}\Big).$$

□

Remark 24 (Full constant chain)**.** The constants are produced in the order:

1. Selection: $C_{\mathrm{sel}}(N, R), q_{\mathrm{sel}}(N, R), \tau_{\mathrm{reg}}(N, R)$ from `quantitative-selection-principle.md` (depending on BDV Lemmas 4.9, 4.12, 4.16).
2. Graph entry: $\alpha_{\mathrm{graph}}(N, R)$ from $\alpha_{\mathrm{graph}} = (h_F/C'_H)^{2N}$, where $h_F = \mu\rho_\mu/16$ and μ, ρ_μ are BDV Theorem 4.18 thresholds.
3. Bootstrap: $C_{\mathrm{boot}}(N, R)$ from Kinderlehrer–Nirenberg / Lieberman applied with BDV Lemma 4.16's L_q.
4. Nearly spherical: $c_{\mathrm{sph}}(N), \delta_{\mathrm{sph}}(N)$ from BDV Theorem 5.1.
5. Asymmetry vs. L^2: $C_{\mathrm{BDV}}(N)$ from BDV Lemma 4.2 + 5.2.
6. Saint–Venant quotient: $c_*(N) = c_{\mathrm{sph}}(N)/C_{\mathrm{BDV}}(N)$.
7. Saint–Venant constant: $q_*(N, R) = c_*(N)/(2C_{\mathrm{sel}}(N, R))$.
8. Global Saint–Venant: $c_{\mathrm{SV}}(N, R) = \min(q_*/C_{\mathrm{Fr}}, c_{\mathrm{far,SV}})$.
9. Kohler–Jobin multiplier: $2L_N/((N+2)T_N)$, universal.
10. Faber–Krahn: $c_{\mathrm{FK}}(N, R) = \min(L_N c_{\mathrm{SV}}/((N+2)T_N), c_{\mathrm{far,FK}}/4)$.

Every constant in this chain depends only on (N, R), and the dependence is through named BDV regularity constants. No constant arises from a limit or compactness argument that hides its (N, R)-dependence; the only compactness-derived constants are $c_{\mathrm{far,SV}}$ and $c_{\mathrm{far,FK}}$, which are far from the ball and do not enter the small-asymmetry regime.

10 Where the contradiction was, and what replaced it

Step	BDV	Plan 1 / Route A
Selection	sequential + compactness limit (contradiction)	single-set + explicit $C_{\mathrm{sel}}(N, R)$
Limit existence	qualitative, $U_n \to U_\infty$ in L^1	not needed
$\alpha \to$ graph	qualitative from limit	explicit $\alpha_{\mathrm{graph}}(N, R)$
Bootstrap	implicit in BDV regularity	explicit Schauder (Kinderlehrer–Niren
Nearly spherical	BDV Theorem 5.1	unchanged
Quotient lower bound	contradiction with $\alpha(U_\infty) > 0$	direct, $Q_\alpha \geq c_*(N)$
Saint–Venant	by contradiction	by computation, $Q_\alpha \geq c_*/2C_{\mathrm{sel}}$
Faber–Krahn transfer	Kohler–Jobin	unchanged

The single contradiction step that was removed. BDV's argument is: "Assume sharp Saint–Venant fails ⇒ minimizing sequence ⇒ compactness limit ⇒ contradiction with nearly spherical."

Plan 1's argument is: "Given any Ω_0 with $\alpha(\Omega_0) \leq \alpha_{\mathrm{graph}}$:

- Construct $\widetilde{\Omega}$ in one step (Thm. 2).
- Note $\widetilde{\Omega}$ is a $C^{2,\gamma}$ graph (Thms. 8, Prop. 9).
- Apply BDV Theorem 5.1 pointwise to read off $Q_\alpha(\widetilde{\Omega}) \geq c_*(N)$.
- Conclude $Q_\alpha(\Omega_0) \geq c_*(N)/(2C_{\mathrm{sel}}(N, R))$ by the selection quotient bound."

No assumption on smallness of $Q_\alpha(\Omega_0)$ is made; the conclusion is a direct lower bound.

11 Status

What is fully written.

- Theorem 2 (selection): full proof in `quantitative-selection-principle.md`.
- Lemma 4 (volume normalization): full proof in `corrections-response.tex` §1.
- Lemmas 5, Props. 6, 7, Thm. 8 (graph entry): full proofs in `graph-entry-closure.md`.
- Prop. 9 ($C^{1,\gamma} \to C^{2,\gamma}$ bootstrap): proof via Kinderlehrer–Nirenberg in `corrections-response.tex` §2.
- Theorem 11: BDV Theorem 5.1, used as black box.
- Lemmas 12, 20, 21: elementary, written in this note.
- Theorems 14, 16, 23: assembled in this note.

What uses BDV as a black box. BDV Lemmas 4.9 (density), 4.12 (Lipschitz/nondegeneracy), 4.16 (Bernoulli coefficient), 4.18 (Alt–Caffarelli flatness improvement), 5.1 (nearly spherical), 5.2 (α vs L^2). All cited by theorem number and used without modification.

What is not numerically tracked. The explicit values of $C_{\mathrm{sel}}, q_{\mathrm{sel}}, \alpha_{\mathrm{graph}}, C_{\mathrm{boot}}, c_{\mathrm{sph}}, C_{\mathrm{BDV}}, c_{\mathrm{far,SV}}, c_{\mathrm{far,FK}}$ are not computed. The point of Route A is the *structural* quantification: every constant is named and (N, R)-trackable, but extracting concrete numbers requires going through each BDV proof.

What this delivers over BDV.

1. BDV's contradiction step (sequential selection + compactness limit) is replaced by Theorem 2, a one-step construction with explicit constants.
2. The graph-entry threshold $\alpha_{\mathrm{graph}}(N, R)$ is given by a formula.
3. The constant $c_{\mathrm{FK}}(N, R)$ is the minimum of a chain of named quantities, traceable to the BDV regularity package.
4. The proof is amenable to numerical/effective tracking: one can take each BDV proof and extract explicit constants, then plug through this chain.

Auditability. Every lemma in this note is either:

- proved in a referenced file with the same notation,
- proved elementarily in this note (Lemmas 12, 20, 21, far-from-ball compactness Lemma 17), or
- quoted from BDV by theorem number (Thms. 4.9, 4.12, 4.16, 4.18, 5.1, 5.2 and Lemma 4.2).

There are no appeals to "a standard argument" that is not pinned to a reference.